1. Introduction

With the rapid growth of online hotel booking platforms, customers increasingly rely on reviews to make decisions about hotels. These reviews contain valuable information about different aspects of a hotel, such as room quality, cleanliness, staff behavior, breakfast, location, facilities, comfort, and value for money. However, hotel reviews are usually written as free-form text, making it difficult to manually identify all the important aspects and understand the sentiment associated with them.

Aspect-Based Sentiment Analysis (ABSA) is a Natural Language Processing (NLP) technique that provides a more detailed analysis of reviews by identifying specific aspects mentioned in the text and determining the sentiment associated with the review. Unlike traditional sentiment analysis, which assigns one overall sentiment to an entire review, ABSA can identify multiple aspects within the same review and analyze them separately.

This project, "Aspect Based Sentiment Analysis on Hotel Reviews," develops a machine learning-based system for analyzing hotel reviews. The project uses the HRAST (Hotel Review Aspect-Based Sentiment Analysis) dataset, which contains hotel reviews annotated with multiple hotel-related aspects and sentiment labels. The dataset contains 23,105 unique reviews and 21 aspects after preprocessing and duplicate removal.

The proposed system first cleans and preprocesses the review text and converts the textual data into numerical features using TF-IDF (Term Frequency–Inverse Document Frequency) with unigram and bigram features. A One-vs-Rest Linear Support Vector Machine (Linear SVM) is then used for multi-label aspect detection, allowing a single review to contain multiple aspects. A Logistic Regression classifier is used for overall sentiment classification into positive, negative, and neutral categories. Since the original sentiment labels are sentence-level rather than aspect-specific, the system further extracts the local context surrounding each detected aspect and uses the sentiment classifier to infer the sentiment associated with that aspect.

The final system therefore provides a structured analysis of hotel reviews by identifying the aspects discussed, determining the overall sentiment, and inferring aspect-level sentiment from the relevant textual context. This can help transform large collections of unstructured hotel reviews into useful information for customers and hotel management.

2. Problem Statement and Objectives

2.1 Problem Statement

Online hotel reviews contain detailed opinions about different aspects of a hotel, but these opinions are presented as unstructured text. Traditional sentiment analysis generally assigns a single sentiment to an entire review and therefore cannot clearly determine **which hotel aspect the sentiment refers to**.

For example, a review such as:

"The room was clean and comfortable, but the staff were unfriendly."

contains multiple aspects with different opinions. The room has a positive sentiment, while the staff have a negative sentiment. A simple overall sentiment classifier may classify the review as positive or negative, but it does not provide this detailed aspect-wise information.

Therefore, there is a need for a machine learning-based system that can:

- identify multiple hotel aspects mentioned in a review,
- classify the overall sentiment of the review,
- and infer the sentiment associated with each detected aspect using its local textual context.

The proposed project addresses this problem using **TF-IDF features, multi-label Linear SVM, Logistic Regression, and local-context sentiment inference**.

2.2 Objectives

The main objectives of this project are:

1. **To preprocess hotel review text** by performing text cleaning, normalization, and removal of unnecessary characters.
2. **To analyze the hotel review dataset** and understand its sentiment distribution, aspect distribution, text characteristics, and relationships between different aspects.
3. **To represent review text numerically** using TF-IDF features with unigram and bigram representations.
4. **To detect multiple hotel aspects** from a single review using a multi-label classification approach based on One-vs-Rest Linear SVM.
5. **To classify the overall sentiment** of each review into positive, negative, or neutral categories using Logistic Regression.
6. **To handle the imbalanced neutral sentiment class** by tuning the decision threshold for the neutral category.
7. **To extract the local textual context** surrounding each detected aspect in order to identify the portion of the review relevant to that aspect.
8. **To infer aspect-level sentiment** from the extracted local context using the trained sentiment classifier.
9. **To evaluate the system** using appropriate performance measures such as Accuracy, Precision, Recall, F1-score, Micro F1, Macro F1, and Hamming Loss.
10. **To develop an end-to-end ABSA pipeline** that converts unstructured hotel reviews into structured information about hotel aspects and their associated sentiments.

3. Methodology

The proposed system performs **Aspect-Based Sentiment Analysis (ABSA)** on hotel reviews. The methodology consists of data preparation, exploratory analysis, text preprocessing, aspect detection, sentiment classification, model selection, aspect-context analysis, and final ABSA prediction.

3.1 Dataset

The project uses the **HRAST hotel review dataset**.

After data cleaning and removal of exact duplicate reviews, the final modeling dataset contains:

- **23,105 hotel reviews**
- **21 hotel-related aspects**
- Three sentiment classes:
    - Positive
    - Negative
    - Neutral

The 21 aspects are:

1. Clean
2. Comfort
3. Facilities/Amenities
4. Location
5. Restaurant (dinner)
6. Staff
7. View (Balcony)
8. Breakfast
9. Room
10. Pool
11. Beach
12. Bathroom/Shower (toilet)
13. Bar
14. Bed
15. Parking
16. Noise
17. Reception-checkin
18. Lift
19. Value for money
20. Wi-Fi
21. Generic

The dataset contains **multi-label aspect annotations**, meaning one review can contain multiple aspects.

3.2 Data Cleaning

The raw dataset was first examined for missing values, duplicate records, inconsistent data types, and anomalous values.

The unnecessary Unnamed: 2 column was removed because it contained almost entirely missing values.

The Location aspect contained an anomalous non-numeric value (σ). This value was handled during preprocessing so that the aspect column could be represented consistently as a binary variable.

All 21 aspect columns were converted into binary labels:

0 → aspect not present

1 → aspect present

Exact duplicate reviews were removed to prevent repeated textual samples from appearing in the modeling dataset.

After cleaning:

23,105 unique reviews were used for modeling.

3.3 Exploratory Data Analysis

Exploratory Data Analysis was performed to understand the characteristics of the dataset.

The analysis included:

- Missing-value analysis
- Duplicate analysis
- Sentiment distribution
- Aspect distribution
- Aspect co-occurrence
- Review length analysis
- Word-frequency analysis

The dataset contains approximately:

- **51.14% Positive**
- **45.45% Negative**
- **3.44% Neutral**

This shows a significant class imbalance, particularly for the Neutral class.

The analysis also showed that approximately **41.84% of reviews contain multiple aspects**, confirming that aspect detection should be treated as a **multi-label classification problem**.

The average review contains approximately **10.3 words**.

3.4 Text Preprocessing

The review text was normalized before feature extraction.

The preprocessing pipeline included:

1. Converting text to lowercase
2. Removing unnecessary special characters
3. Normalizing whitespace
4. Removing leading and trailing spaces
5. Preserving useful textual information such as negation words

For example:

"The ROOM was very CLEAN!"

becomes approximately:

"the room was very clean"

A processed text representation was then used as input to the machine-learning models.

3.5 Train, Validation and Test Split

Two separate modeling tasks were created.

Aspect Detection

The aspect task used:

- Training: **16,173 reviews**
- Validation: **3,466 reviews**
- Test: **3,466 reviews**

Since a review can contain multiple aspects, the original 21 binary aspect labels were retained.

Sentiment Classification

The four reviews containing multiple sentiment labels were treated as ambiguous and excluded from sentiment training/evaluation.

This resulted in **23,101 sentiment samples**.

The sentiment data was split using **stratification** to preserve the Positive, Negative and Neutral proportions.

3.6 TF-IDF Feature Extraction

The textual reviews were converted into numerical feature vectors using **TF-IDF — Term Frequency-Inverse Document Frequency**.

The configuration used was:

Maximum features = 10,000

N-gram range = 1–2

Minimum document frequency = 2

Sublinear TF = True

Therefore, the model considers both:

- **Unigrams** — individual words
- **Bigrams** — two-word combinations

For example:

"very clean room"

can generate features such as:

very

clean

room

very clean

clean room

The TF-IDF vectorizer was fitted only on the training data and then used to transform validation and test data. This prevents information from the validation or test sets from influencing feature learning.

3.7 Aspect Detection

Aspect detection is formulated as a **multi-label classification problem** because a single review can mention multiple hotel aspects.

For example:

"The room was clean and comfortable."

contains:

Room = 1

Clean = 1

Comfort = 1

while the other aspects are 0.

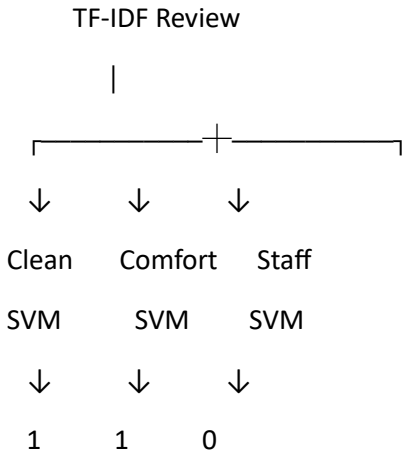
Several models were evaluated during development.

The final selected model was:

One-vs-Rest Linear SVM

One-vs-Rest trains an individual binary classifier for each aspect.

Conceptually:



The final aspect model achieved on the unseen test set:

Metric	**Score**
Micro F1	**0.8525**
Macro F1	**0.8451**
Precision	**0.8354**
Recall	**0.8703**
Hamming Loss	**0.0231**

3.8 Sentiment Classification

Sentiment classification was treated as a three-class classification problem:

Positive

Negative

Neutral

Several models were compared:

- Logistic Regression
- Balanced Linear SVM
- Unbalanced Linear SVM

Although Linear SVM achieved higher accuracy, Logistic Regression provided the best **Macro F1** and the best performance for the minority Neutral class during validation.

Therefore, the final sentiment model was:

Logistic Regression

The final test performance was:

Metric	Score
Accuracy	**0.8904**
Macro F1	**0.6943**
Weighted F1	**0.8981**

3.9 Neutral-Class Threshold Tuning

The Neutral class was highly underrepresented.

Therefore, simply selecting the class with the highest probability resulted in poor Neutral detection.

The Logistic Regression model provides probability estimates for:

P(Negative)

P(Neutral)

P(Positive)

Different Neutral thresholds were evaluated on the **validation set**.

The best threshold was:

Neutral threshold = 0.44

This improved validation performance:

Original Macro F1 → 0.7106

Tuned Macro F1 → 0.7142

Original Neutral F1 → 0.3046

Tuned Neutral F1 → 0.3129

The selected threshold was then kept fixed when evaluating the final model on the test set.

3.10 Aspect Context Extraction

A major limitation of the dataset is that sentiment is provided at the **sentence level**, rather than separately for each aspect.

For example:

"The room was clean and comfortable, but the staff were unfriendly."

The dataset does not directly provide:

Room → Positive

Clean → Positive

Comfort → Positive

Staff → Negative

Therefore, an additional **aspect-context extraction layer** was implemented.

The system identifies the local sentence or clause associated with each detected aspect.

For example:

Room

→ "the room was clean and comfortable"

Staff

→ "the staff were unfriendly"

The sentiment classifier is then applied to these local contexts.

3.11 Aspect-Level Sentiment Inference

The local context extracted for each detected aspect is converted into TF-IDF features and passed through the trained Logistic Regression model.

The final process is:

Detected Aspect

↓

Find Local Context

↓

TF-IDF

↓

Logistic Regression

↓

Positive / Negative / Neutral

For example:

Input:

"The room was clean and comfortable, but the staff were unfriendly."

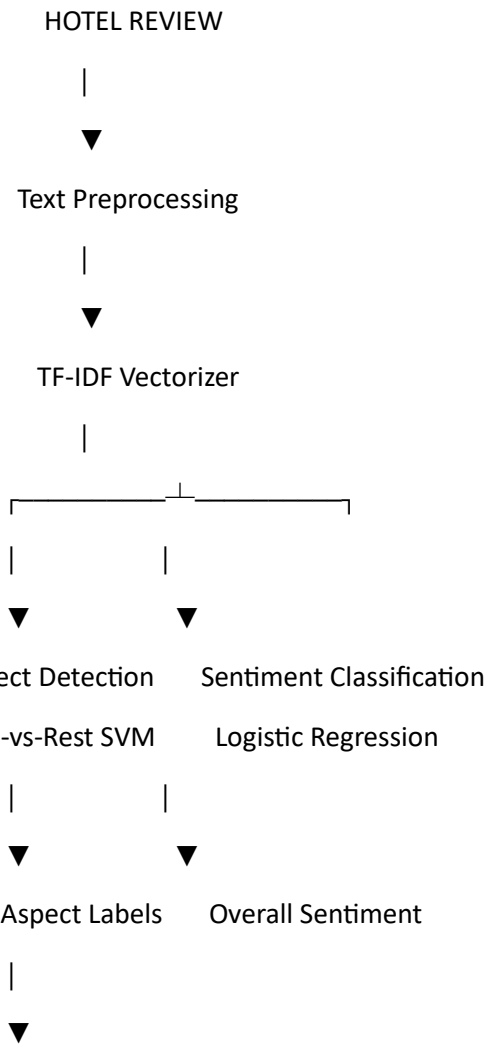
The system produces:

Aspect	Inferred Sentiment
Clean	Positive
Comfort	Positive
Room	Positive
Staff	Negative

This allows the system to represent different sentiments for different aspects of the same review.

3.12 Final ABSA Pipeline

The complete system operates as follows:



Aspect Context Extraction

|

▼

Local Sentiment Analysis

|

▼

Aspect-Level Sentiment

|

▼

FINAL ABSA OUTPUT

3.13 Final Prediction Example

For:

"The room was clean and comfortable, but the staff were unfriendly."

the system detects:

Clean

Comfort

Staff

Room

and generates:

Clean → Positive

Comfort → Positive

Room → Positive

Staff → Negative

This demonstrates the main purpose of ABSA: identifying **what aspect is being discussed and the sentiment associated with that aspect**.

3.14 Final Results

Aspect Detection

One-vs-Rest Linear SVM

- Micro F1: **85.25%**

- Macro F1: **84.51%**
- Precision: **83.54%**
- Recall: **87.03%**

Sentiment Classification

Logistic Regression

- Accuracy: **89.04%**
- Macro F1: **69.43%**
- Weighted F1: **89.81%**

The strongest aspect detection performance was obtained for:

- Pool: **97.71% F1**
- Parking: **95.95%**
- Wi-Fi: **95.35%**
- Breakfast: **94.88%**
- Location: **94.46%**

The weakest aspects were:

- Generic: **58.26%**
- Restaurant (dinner): **65.26%**
- Value for money: **70.55%**

For sentiment classification:

- Positive F1: **93.65%**
- Negative F1: **90.44%**
- Neutral F1: **24.20%**

The low Neutral F1 is mainly associated with the severe class imbalance in the dataset.

4. Results and Discussion

We should include:

4.1 Final Model Performance

Use your final table:

Task	Model	Test Result
Aspect Detection	One-vs-Rest Linear SVM	Micro F1 = **0.8525**
Aspect Detection	One-vs-Rest Linear SVM	Macro F1 = **0.8451**

Task	Model	Test Result
Sentiment Classification	Logistic Regression	Accuracy = **0.8904**
Sentiment Classification	Logistic Regression	Macro F1 = **0.6943**
Sentiment Classification	Logistic Regression	Weighted F1 = **0.8981**

4.2 Aspect Detection Results

Insert your **Aspect Detection Performance by Aspect** chart.

Then explain the strongest and weakest aspects.

4.3 Sentiment Classification Results

Insert your **Sentiment Classification Performance** chart.

Explain:

- Positive F1 = **0.9365**
- Negative F1 = **0.9044**
- Neutral F1 = **0.2420**

4.4 Confusion Matrix

Insert the confusion matrix you already generated.

Explain why Neutral is frequently confused with Positive/Negative.

4.5 Custom ABSA Demonstration

Insert a screenshot of your final working output:

"The room was clean and comfortable, but the staff were unfriendly."

Then show:

Aspect	Sentiment
Clean	Positive
Comfort	Positive
Room	Positive
Staff	Negative

This is one of your strongest project demonstrations.

4.6 Discussion

Explain what the results mean, including:

- Aspect detection performs strongly overall.
- Positive and Negative sentiment are classified well.

- Neutral remains challenging because it is severely underrepresented.
- The system can identify multiple aspects in one review.
- Local context allows different sentiments to be inferred for different aspects.

5. Limitations

The proposed system has the following limitations:

5.1 Sentence-Level Sentiment Labels

The dataset provides sentiment labels at the **sentence/review level**, rather than separate sentiment labels for every aspect. Therefore, the aspect-level sentiments generated by the final system are **inferred from the local context** of each detected aspect rather than directly learned from gold aspect-specific sentiment annotations.

5.2 Class Imbalance

The Neutral sentiment class is significantly smaller than the Positive and Negative classes. This imbalance affects the model's ability to correctly identify neutral reviews.

The final test results demonstrate this clearly:

- Positive F1: **0.9365**
- Negative F1: **0.9044**
- Neutral F1: **0.2420**

5.3 Difficulty with Ambiguous Language

Hotel reviews can contain mixed, indirect, or ambiguous opinions. Expressions such as sarcasm, comparisons, or statements with both positive and negative opinions may be difficult for a traditional TF-IDF-based classifier to interpret.

5.4 Limited Context Understanding

TF-IDF represents text using word and short-phrase frequencies. It does not fully understand deeper semantic relationships, long-range dependencies, or the context-dependent meaning of words.

5.5 Aspect Detection Variability

Some aspects have significantly fewer examples than others. Consequently, the model performs better for frequently represented aspects such as Pool, Parking, Wi-Fi, and Breakfast, while less frequent or more generic aspects such as Generic and Restaurant (dinner) are more difficult to detect.

5.6 Context-Based Aspect Sentiment Inference

The local-context approach improves aspect-level sentiment interpretation, but it depends on correctly identifying the textual context associated with an aspect. Complex sentences containing several clauses or references may still produce incorrect aspect-sentiment associations.

6. Future Scope

The project can be extended in several ways:

6.1 Transformer-Based Models

Future work can investigate transformer-based models such as **BERT, RoBERTa, or domain-adapted language models** to capture deeper semantic relationships and contextual meaning.

6.2 Aspect-Specific Sentiment Annotation

A stronger ABSA dataset could be created by manually annotating each aspect with its own sentiment:

Room → Positive

Clean → Positive

Comfort → Positive

Staff → Negative

This would allow the system to train and evaluate **true aspect-level sentiment classification** instead of inferring sentiment from sentence-level labels.

6.3 Better Handling of Class Imbalance

Future versions could explore additional techniques such as:

- Advanced class-weighting strategies
- Focal loss
- Controlled oversampling
- Data augmentation
- Synthetic minority examples

These approaches could improve Neutral sentiment detection.

6.4 Context-Aware Models

Instead of relying primarily on keyword-based context extraction, future systems could use dependency parsing, attention mechanisms, or transformer-based contextual representations to identify exactly which sentiment expression belongs to each aspect.

6.5 Explainable ABSA

The system could provide explanations showing **which words or phrases contributed to each aspect and sentiment prediction**. This would make the predictions easier for users and hotel managers to understand.

6.6 Real-Time Hotel Review Analysis

The system could be developed into a web-based application where hotel managers can upload or enter new reviews and receive:

- Detected aspects
- Aspect-level sentiment
- Positive/negative aspect summaries

- Overall sentiment statistics
- Frequently mentioned problems

6.7 Hotel Performance Dashboard

A future dashboard could aggregate thousands of reviews and display trends such as:

Cleanliness → 82% Positive

Staff → 91% Positive

Breakfast → 76% Positive

Room → 68% Positive

Wi-Fi → 54% Positive

This could help hotels identify areas requiring improvement.

6.8 Multilingual Hotel Review Analysis

The system could be extended to support reviews written in multiple languages, allowing it to analyze a broader range of international hotel reviews.

7. Conclusion

The project successfully developed an **Aspect-Based Sentiment Analysis system for hotel reviews** using machine-learning techniques. The proposed approach combines multi-label aspect detection with sentiment classification to provide more detailed information than conventional overall sentiment analysis.

The system uses **TF-IDF** for textual feature extraction, **One-vs-Rest Linear SVM** for detecting 21 hotel-related aspects, and **Logistic Regression** for three-class sentiment classification. The Neutral probability threshold was tuned to **0.44** to improve the detection of the minority Neutral class.

On the unseen test set, the aspect detection model achieved a **Micro F1-score of 0.8525** and a **Macro F1-score of 0.8451**, demonstrating strong performance in identifying multiple hotel aspects. The sentiment classifier achieved **89.04% accuracy** and **0.8981 weighted F1-score**.

The final system can identify multiple aspects within a single review and infer sentiment from the local context associated with each aspect. For example, a review containing both positive comments about a room and negative comments about staff can be represented with different sentiments for the respective aspects.

However, the study also identified the difficulty of classifying Neutral sentiment, which achieved an F1-score of **0.2420** on the test set due primarily to the strong class imbalance.

Overall, the project demonstrates that machine-learning-based ABSA can provide useful and interpretable insights from hotel reviews by identifying **what customers are discussing and whether their comments are positive, negative, or neutral**. Future improvements using transformer-based models and aspect-specific sentiment annotations could further improve the system.

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8. References

Use these references in your Word report.

Dataset

[1] HRAST, *Hotel Review Aspect-Based Sentiment Analysis Dataset*, Kaggle.
Available at: https://www.kaggle.com/

TF-IDF

[2] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.

Logistic Regression

[3] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*, 2nd ed. Springer, 2009.

Support Vector Machine

[4] C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, pp. 273–297, 1995.

TF-IDF / Machine Learning implementation

[5] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

Aspect-Based Sentiment Analysis

[6] M. Pontiki et al., "SemEval-2014 Task 4: Aspect Based Sentiment Analysis," *Proceedings of the 8th International Workshop on Semantic Evaluation*, pp. 27–35, 2014.

BERT / Future Scope

[7] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," *Proceedings of NAACL-HLT*, pp. 4171–4186, 2019.

Evaluation Metrics

[8] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, pp. 861–874, 2006.